



EMNLP 2021

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Less Is More : Domain Adaptation with Lottery Ticket for Reading Comprehension

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Reading Comprehension - Task

- Answering questions according to the context
 - Extractive answers are continuous context spans

Context: Victorian schools are either publicly or privately funded. Public schools, also known as state or government schools, are funded and run directly by the Victoria Department of Education. Students do not pay tuition fees, but some extra costs are levied. Private fee-paying schools include parish schools run by the Roman Catholic Church and independent schools similar to British public schools. Independent schools are usually affiliated with Protestant churches. Victoria also has several private Jewish and Islamic primary and secondary schools.

Question: What organization runs the public schools in Victoria?

Answer: Victoria Department of Education

Question: Since students do not pay tuition, what do they have to pay for schooling in Victoria?

Answer: some extra costs

Question: What church runs some private schools in Victoria?

Answer: Roman Catholic Church

Reading Comprehension - SOTA



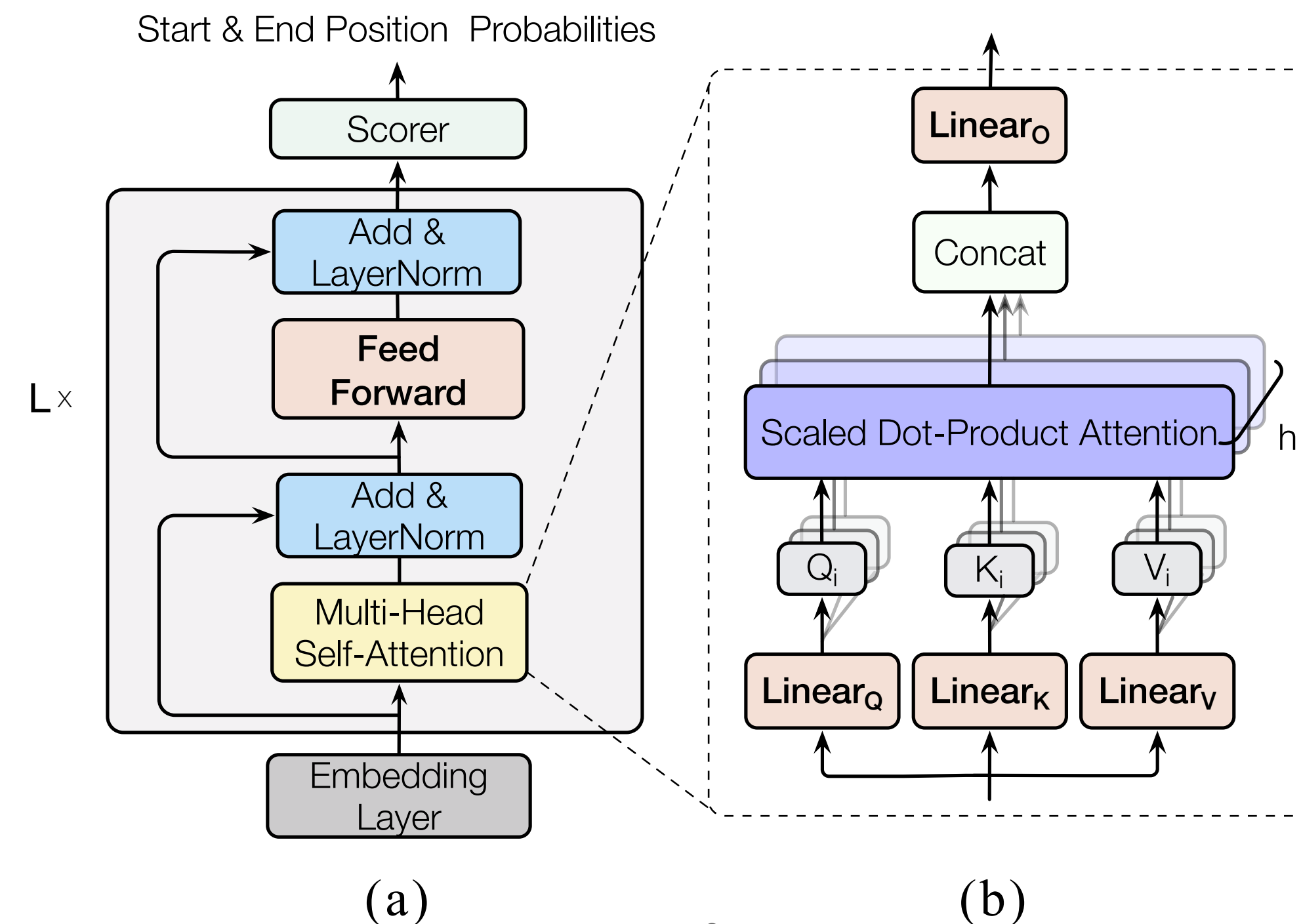
Reading Comprehension - SOTA

- Build on large pre-trained language models with at least **hundreds of millions** parameters



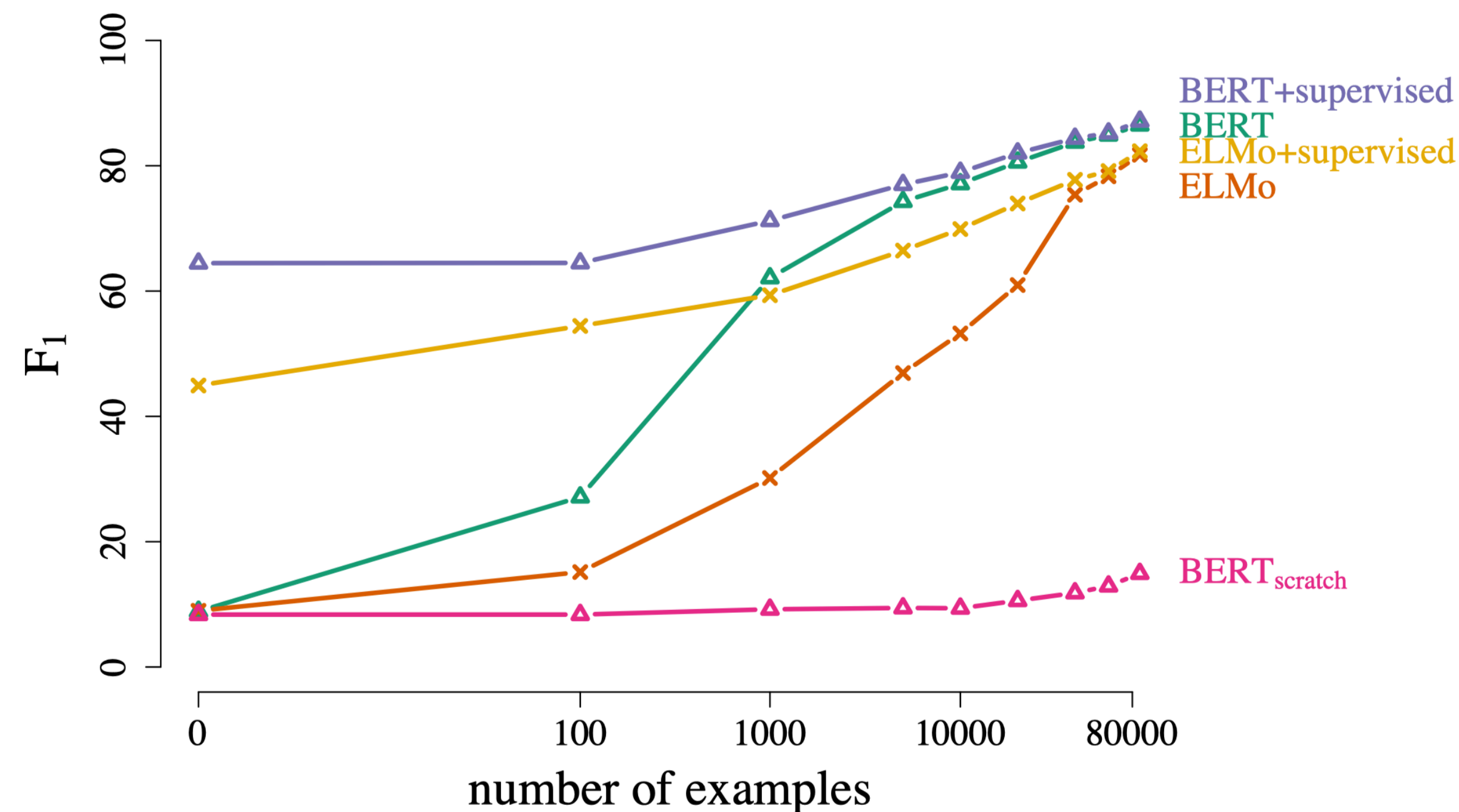
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- Require large amounts of annotated data
 - Generalize poorly

	SQUAD	NEWSQA	SEARCHQA	TQA-G	TQA-W	HOTPOTQA
SQUAD	-	31.8	8.4	37.8	33.4	11.8
NEWSQA	60.4	-	10.1	37.6	28.4	8.0
SEARCHQA	23.3	12.7	-	53.2	35.4	5.2
TQA-G	36.3	18.8	39.2	-	-	8.8
TQA-W	35.5	19.4	27.8	-	-	8.7
HOTPOTQA	54.5	25.6	19.6	37.3	34.9	-
MULTI-75K	-	-	-	-	-	-
SELF	78.0	46.0	52.2	60.7	50.1	24.2



Reading Comprehension - Domain Adaptation



Reading Comprehension - Domain Adaptation

- Unsupervised domain adaptation



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 - Generate synthetic questions (Wang et al., 2019; Shakeri et al., 2020; Rennie et al., 2020)



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- Supervised domain adaptation in few-shot settings
 - **Large** amount of annotated data in source domain 
 - **Limited** annotation in target domain 

Over-parameterized Neural Models



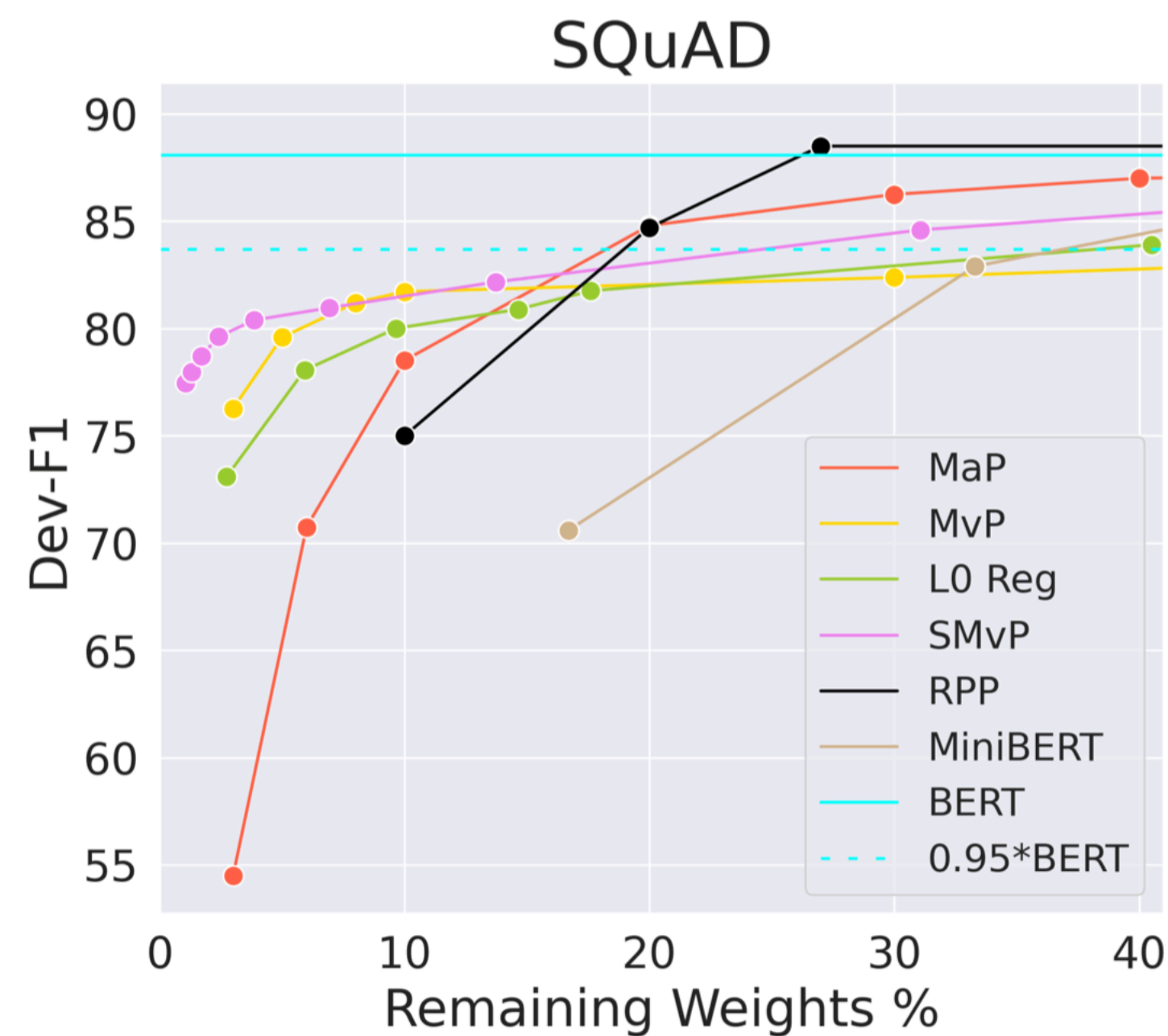
Over-parameterized Neural Models

- Neural networks are over-parameterized, a great fraction of the parameters can be pruned with minimal or even no compromise in performance



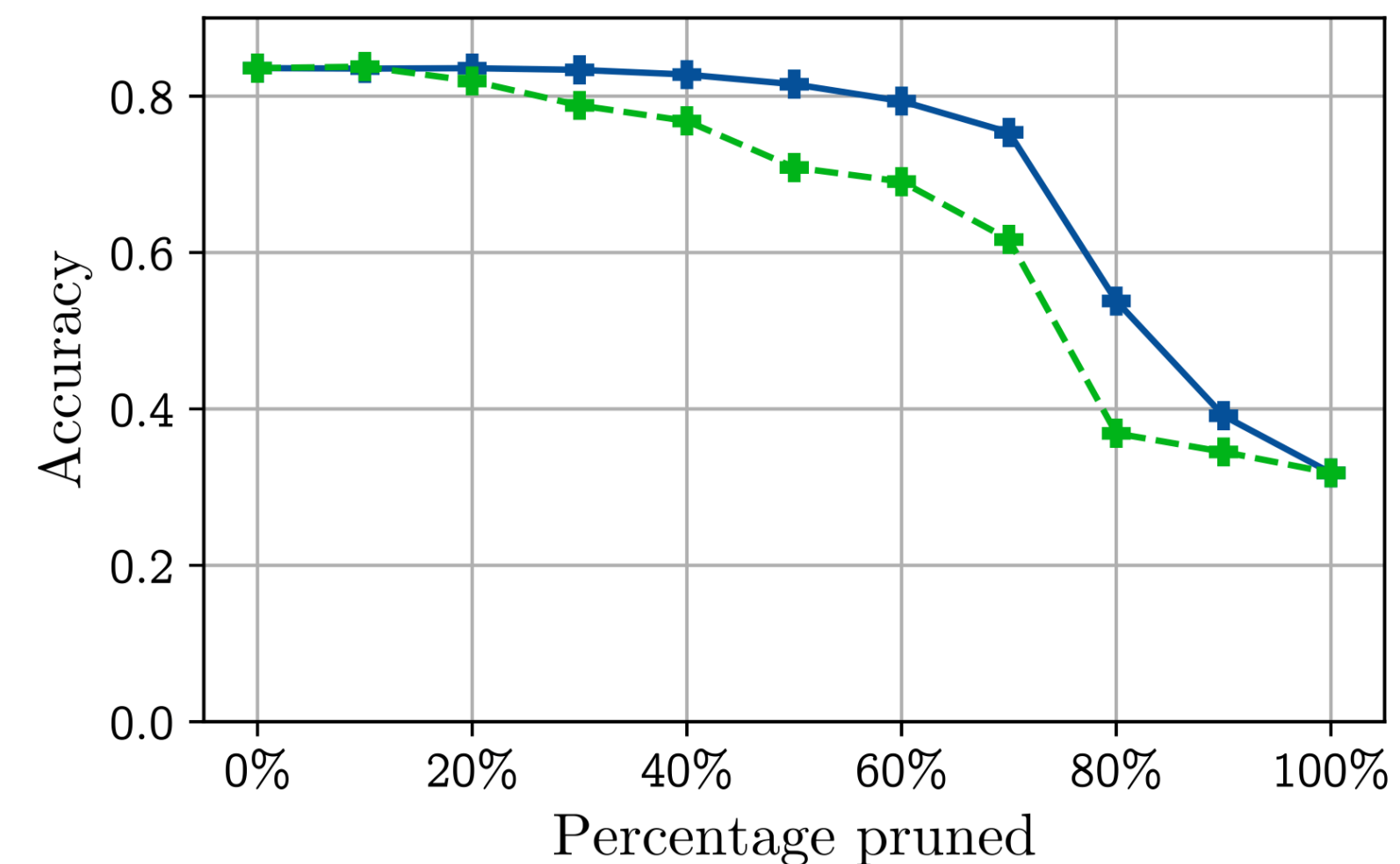
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- **Unstructured** pruning (Frankle et al., 2020; Sanh et al., 2020; Gordon et al., 2020)



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- **Unstructured** pruning (Frankle et al., 2020; Sanh et al., 2020; Gordon et al., 2020)
- **Structured** pruning (Michel et al., 2019; Voita et al., 2019; McCarley et al., 2019; Fan et al., 2020)



(b) Evolution of accuracy on the MultiNLI-matched validation set when heads are pruned from BERT.



Self-Attention Head Importance



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- Self-Attention Attribution (Hao et al. 2021)

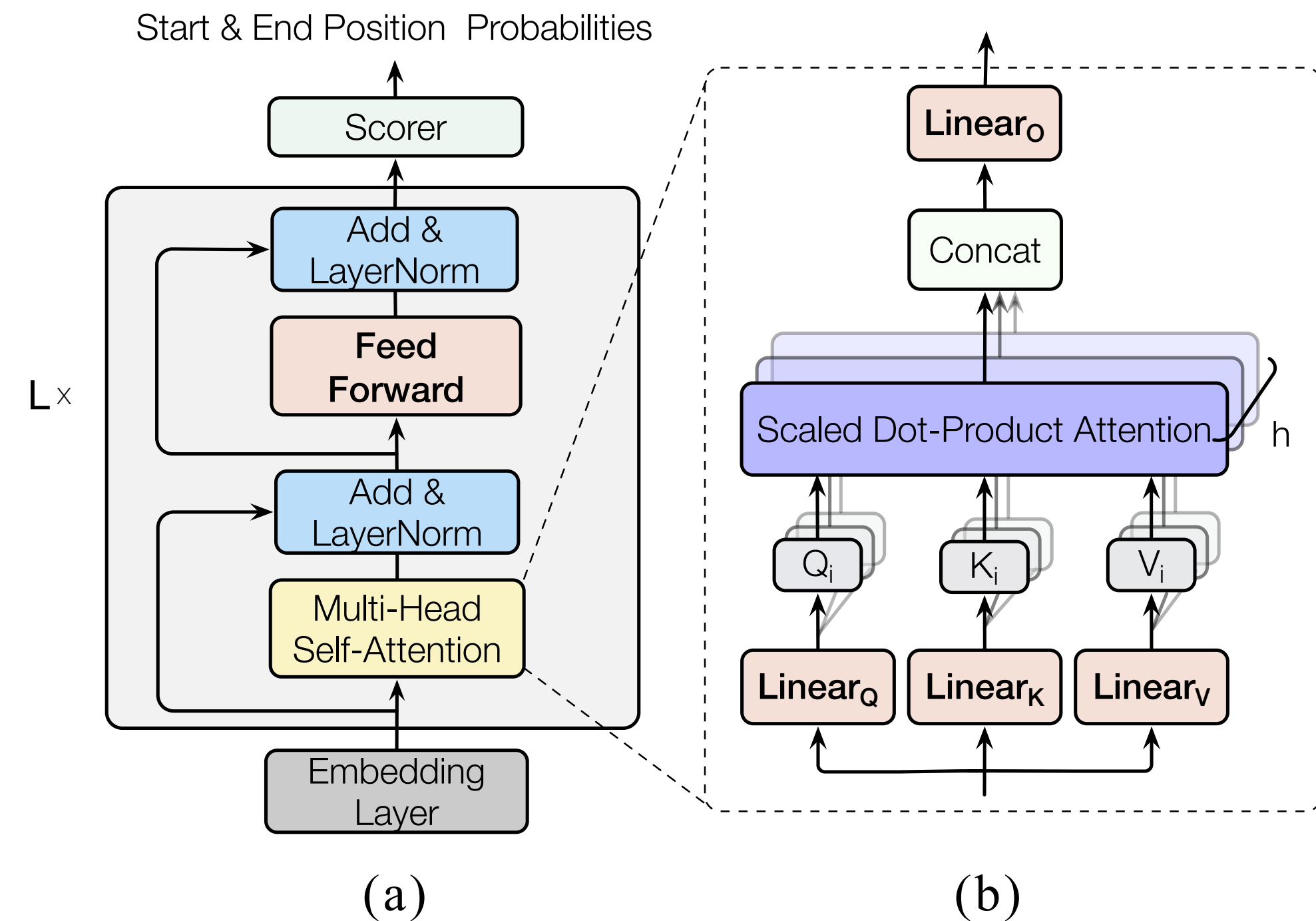


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1. Compute **self-attention map** of i -th head

$$\mathbf{A}_i = \text{softmax}\left(\frac{\mathbf{Q}_i \mathbf{K}_i^\top}{\sqrt{d_k}}\right)$$



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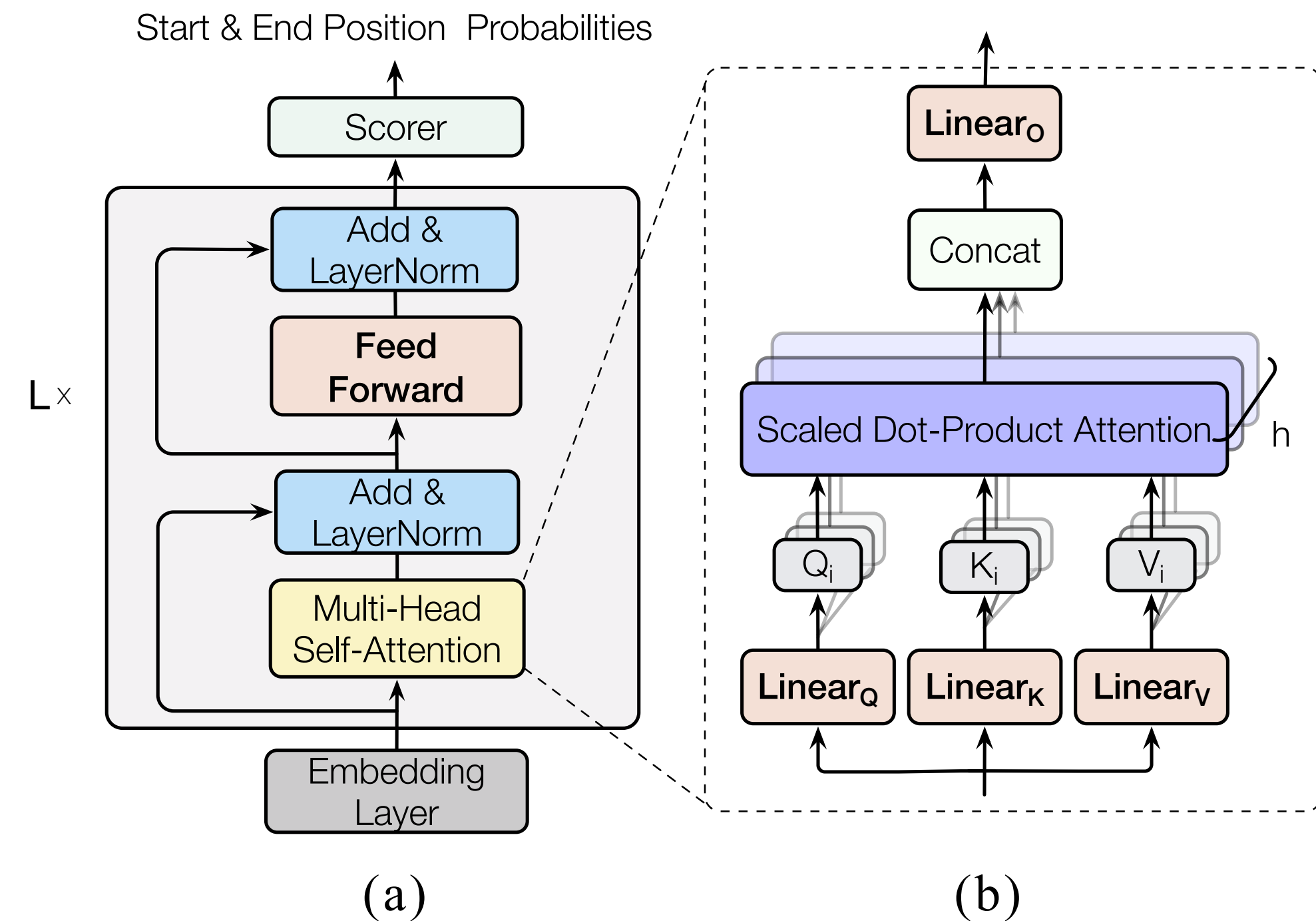
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$$\text{Attr}(\mathbf{A}_i) = \mathbf{A}_i \odot \int_{\alpha=0}^1 \frac{\partial \mathcal{F}(\mathbf{x}, \alpha \mathbf{A})}{\partial \mathbf{A}_i} d\alpha \in \mathbb{R}^{n \times n}$$



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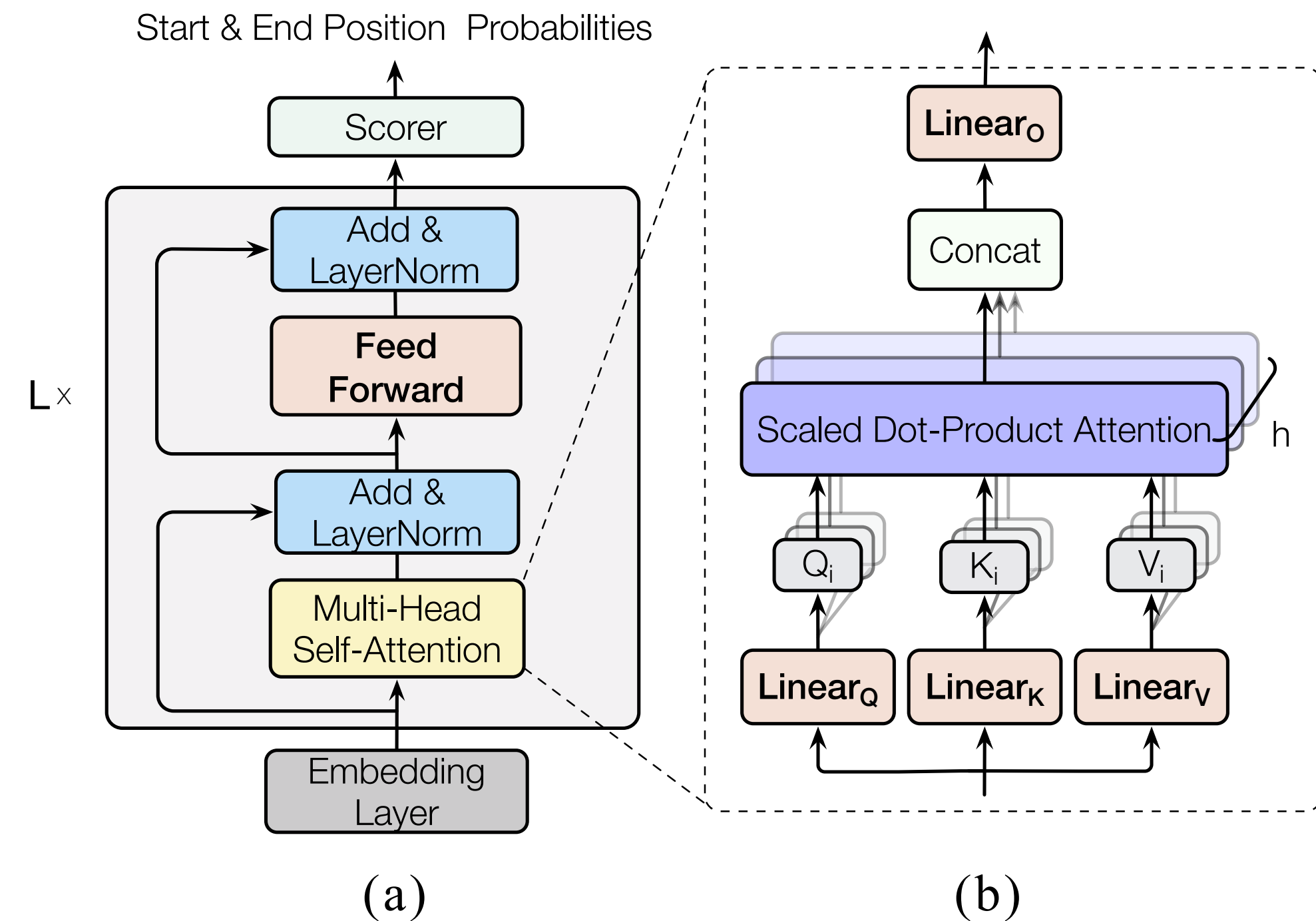
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3. Estimate the **head importance score**

$$I_i = E_{\mathbf{x}} [\max(\text{Attr}(\mathbf{A}_i))]$$



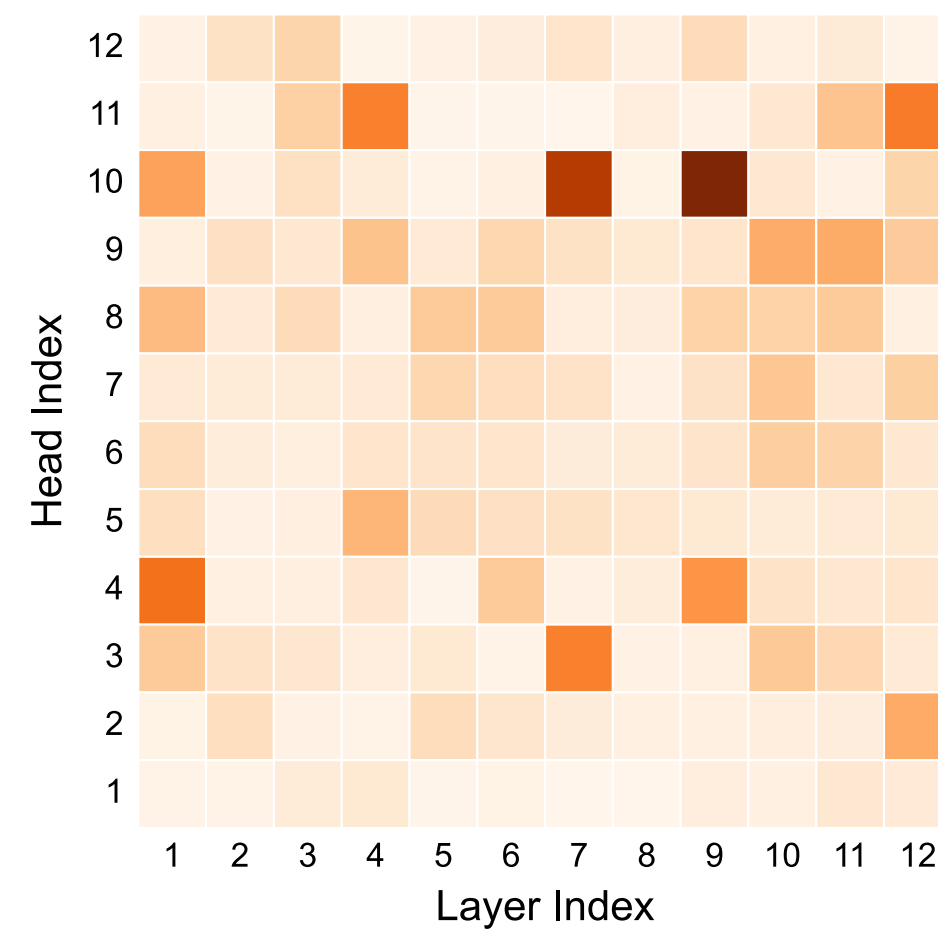
Self-Attention Head Importance Analysis

- Important attention heads are **highly correlated** across various domains
 - e.g., SQuAD, NewsQA, NQ



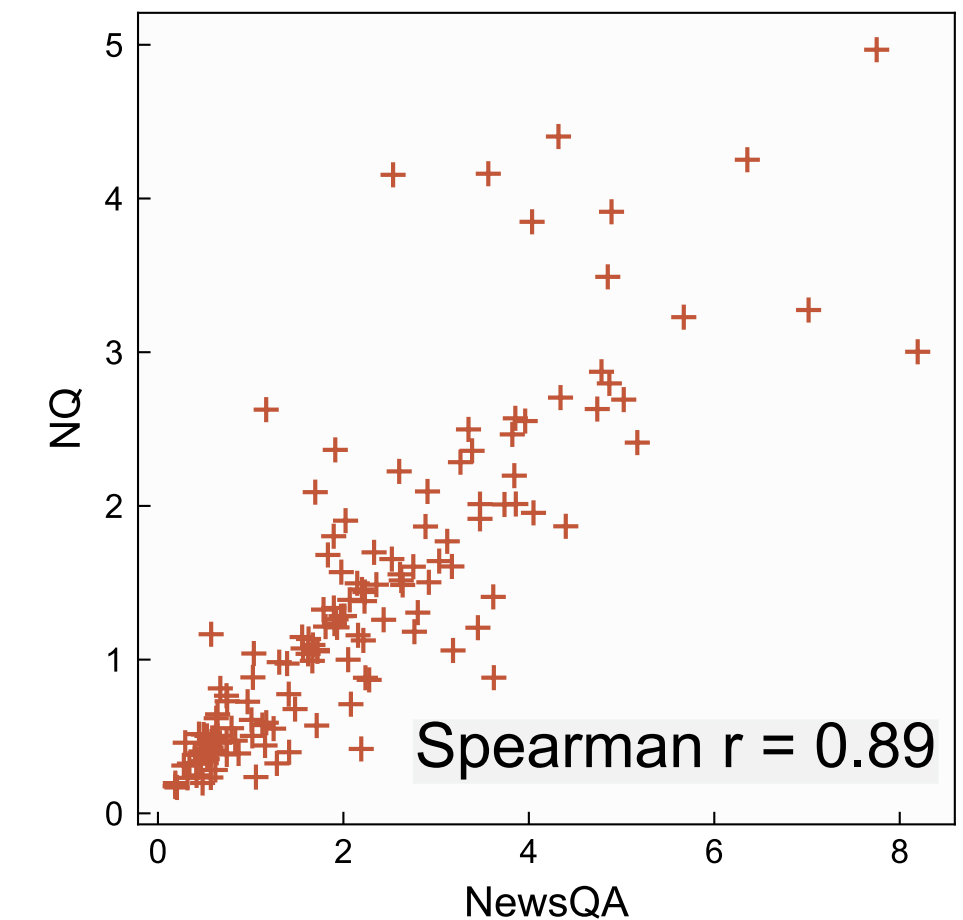
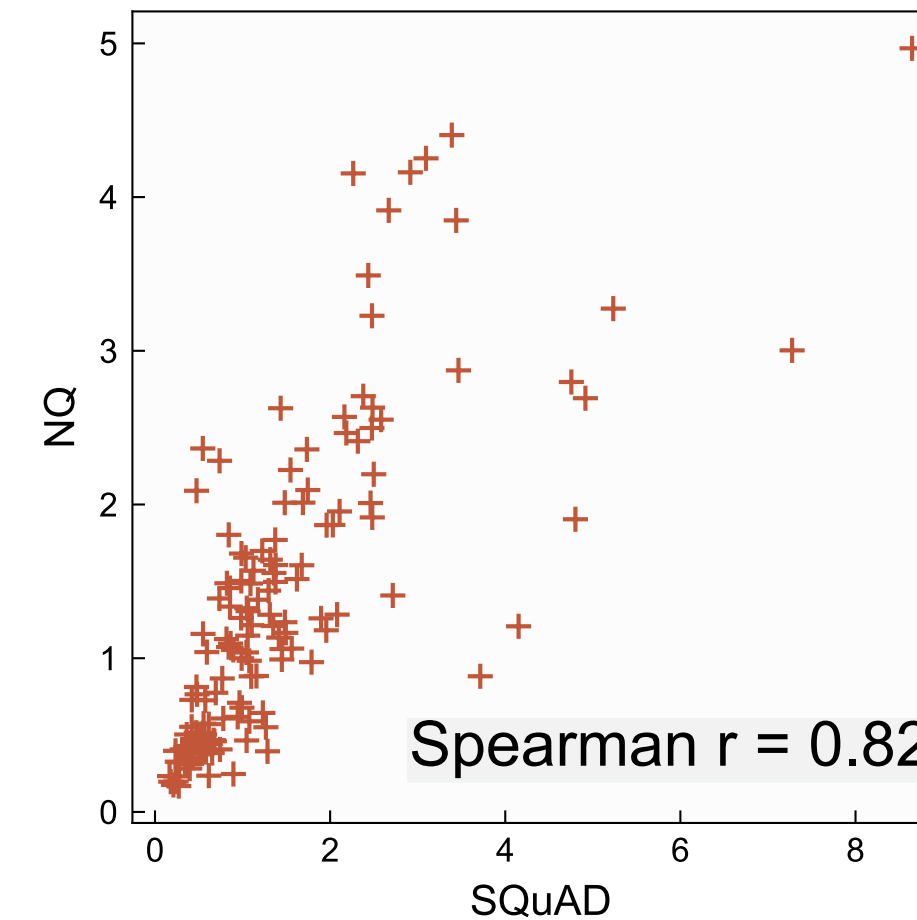
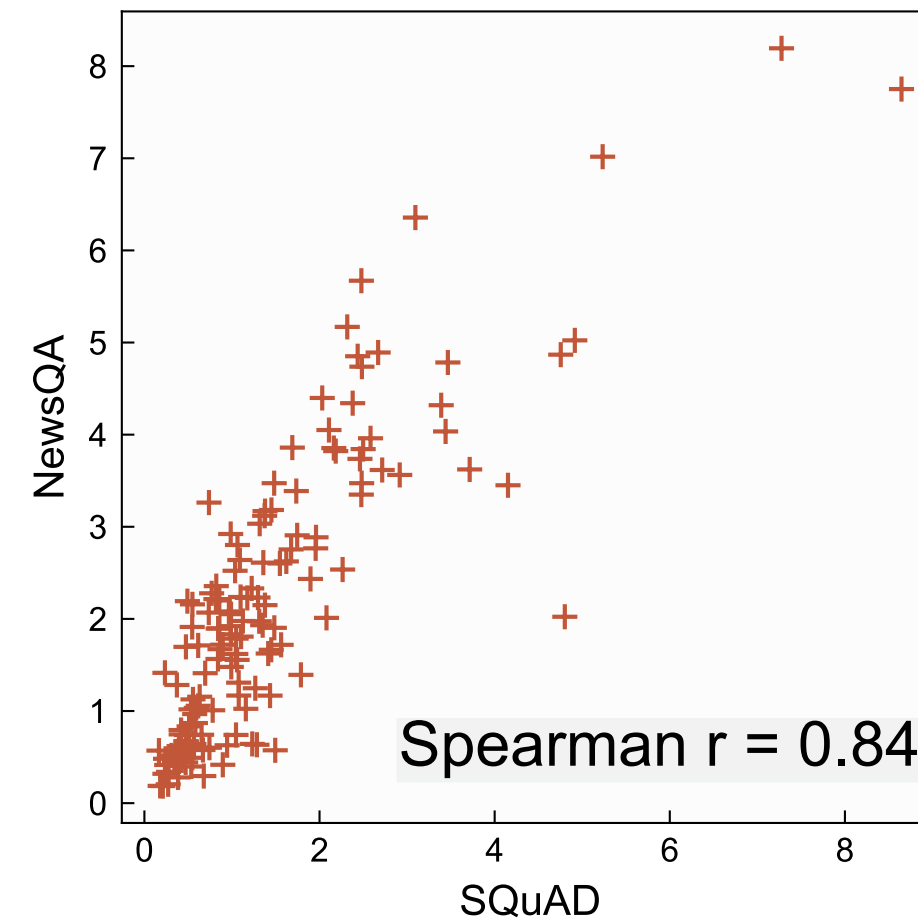
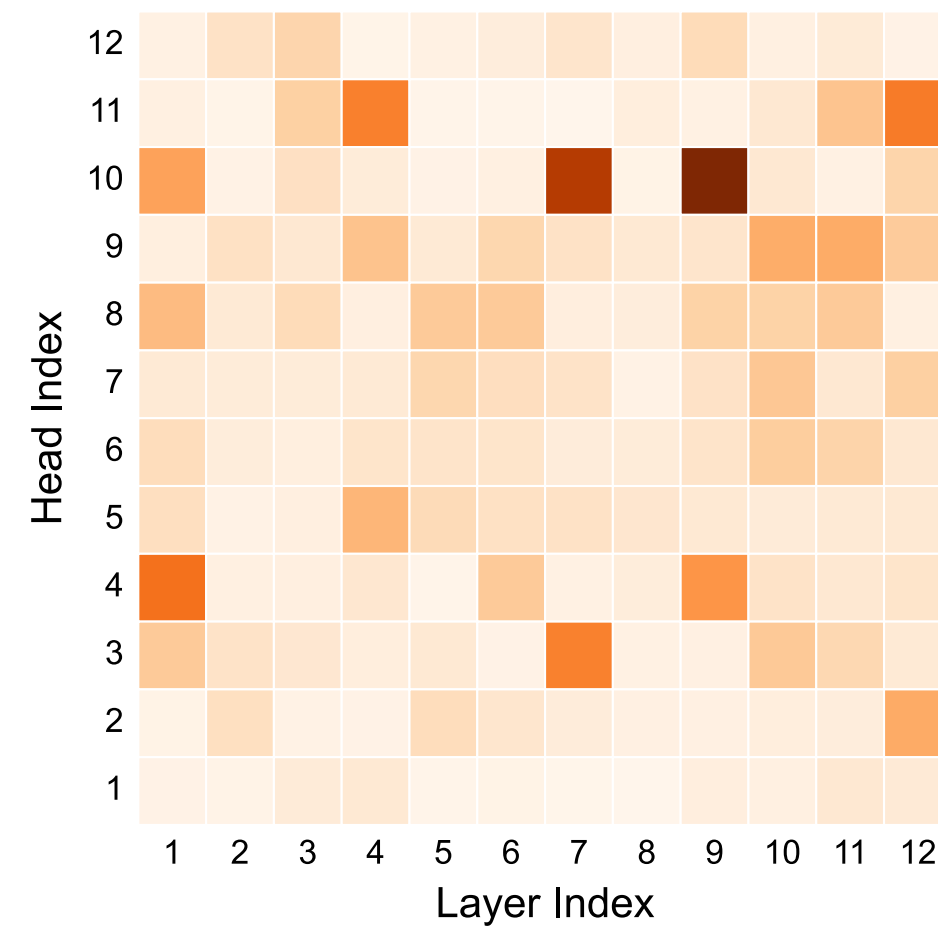
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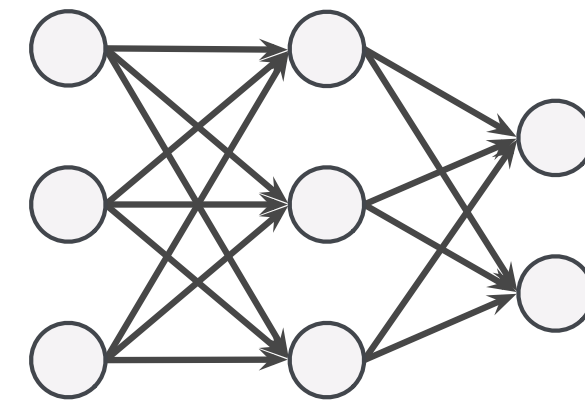


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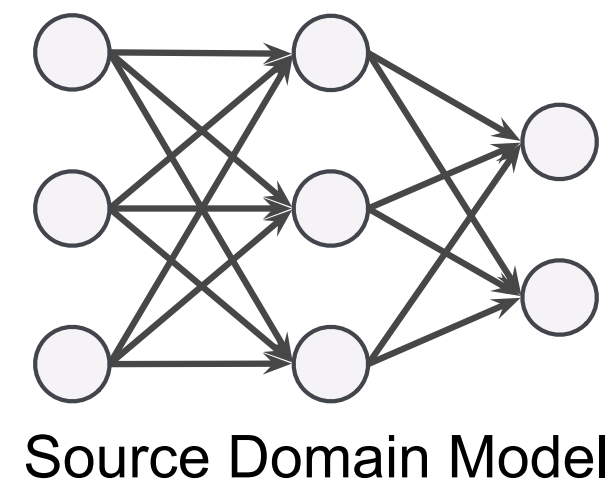
Our Work



Source Domain Model

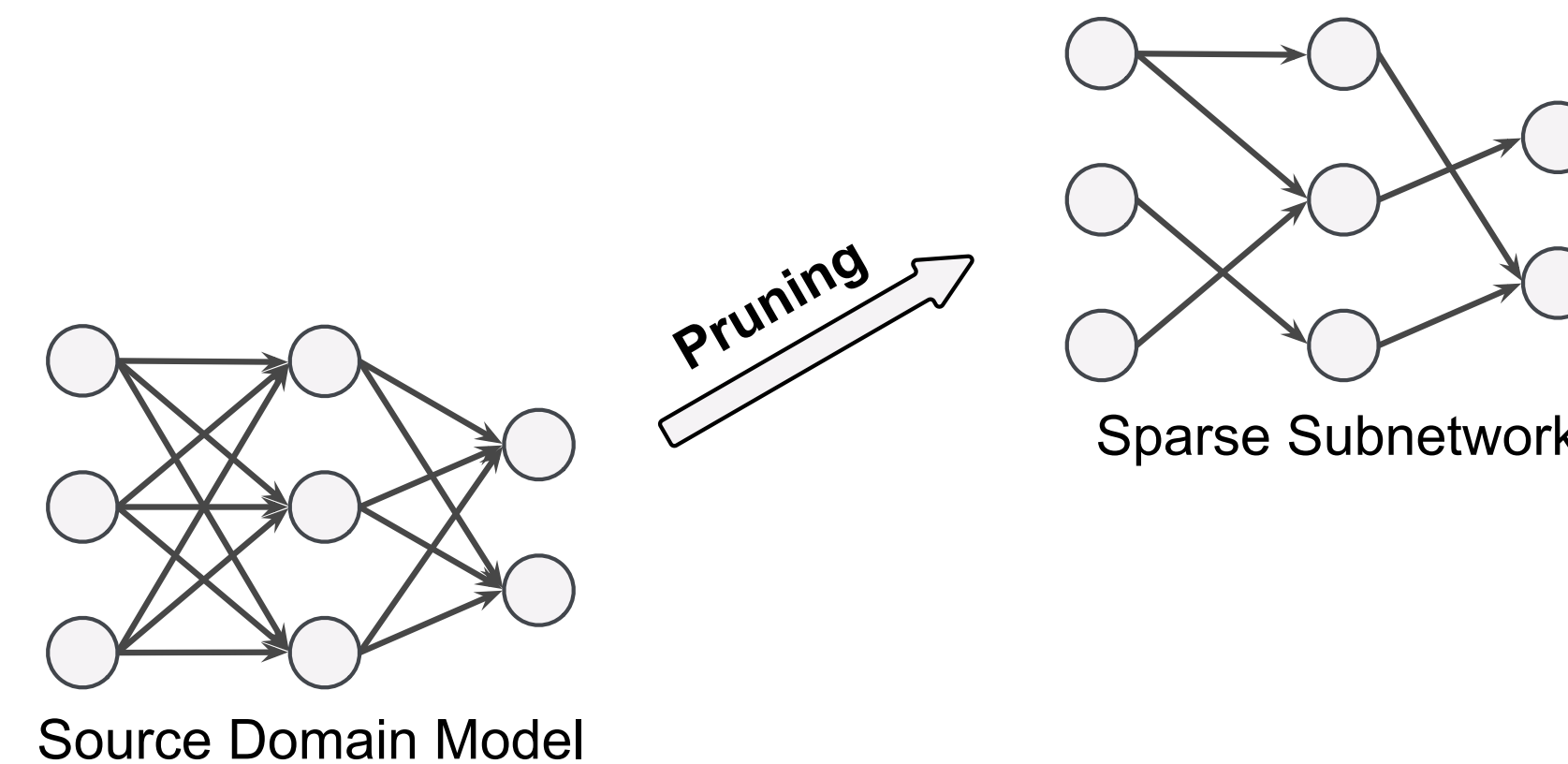
Our Work

- Exploit a small fraction of deliberately selected parameters for adaptation



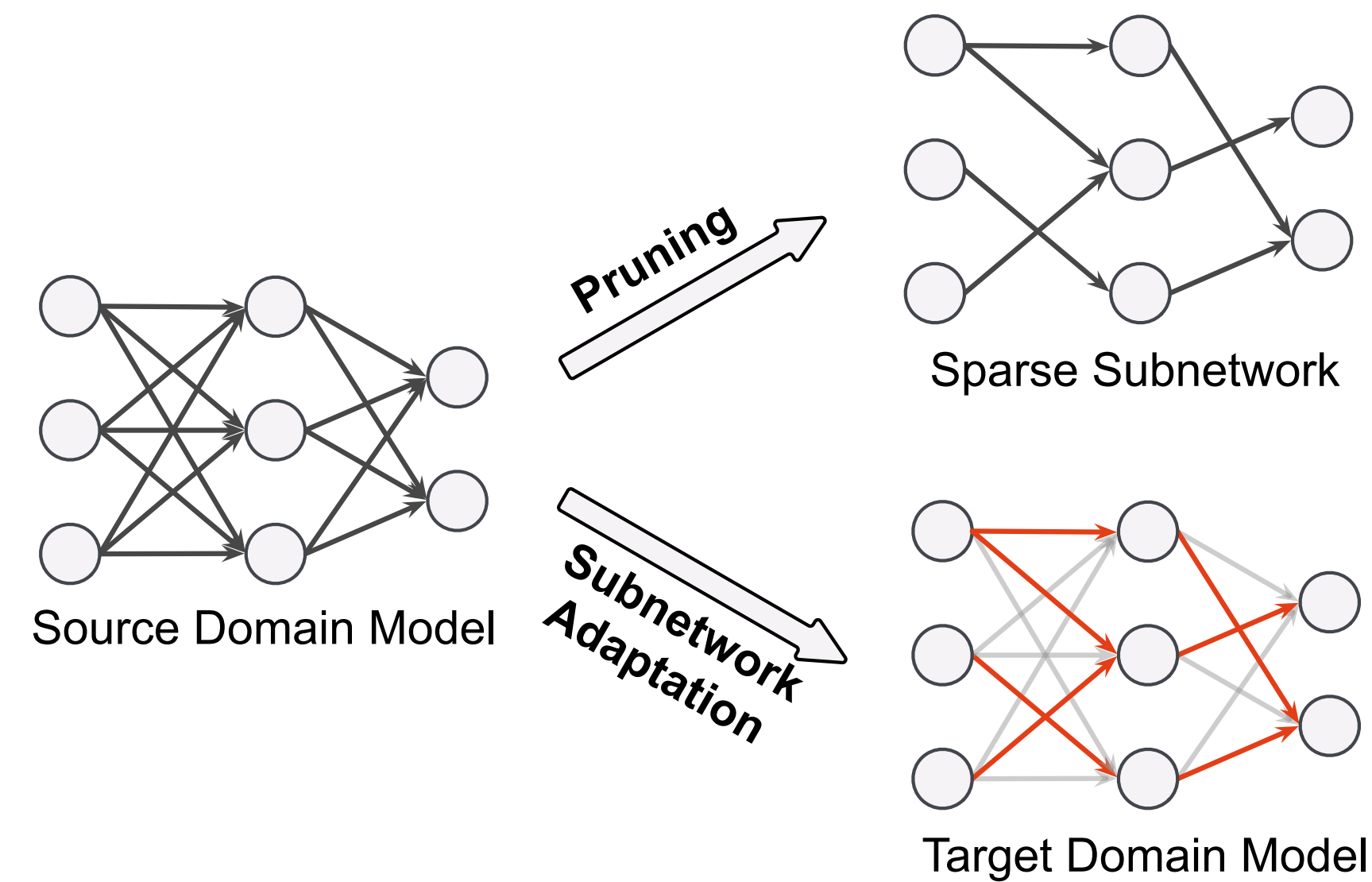
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Our Work

- Exploit a small fraction of deliberately selected parameters for adaptation
 1. Identify **lottery subnetworks** of the source domain model with **self-attention head importance**
 2. Adapt to the target domain by **only updating** the parameters of the **subnetworks**



Identifying the Lottery Network



Identifying the Lottery Network

1. Estimate and normalize attention head importance

$$\hat{I}_n \leftarrow \lambda + (1 - \lambda) \frac{I_n - \min(I_n)}{\max(I_n) - \min(I_n)}$$

λ indicates the intensity of head importance intervention



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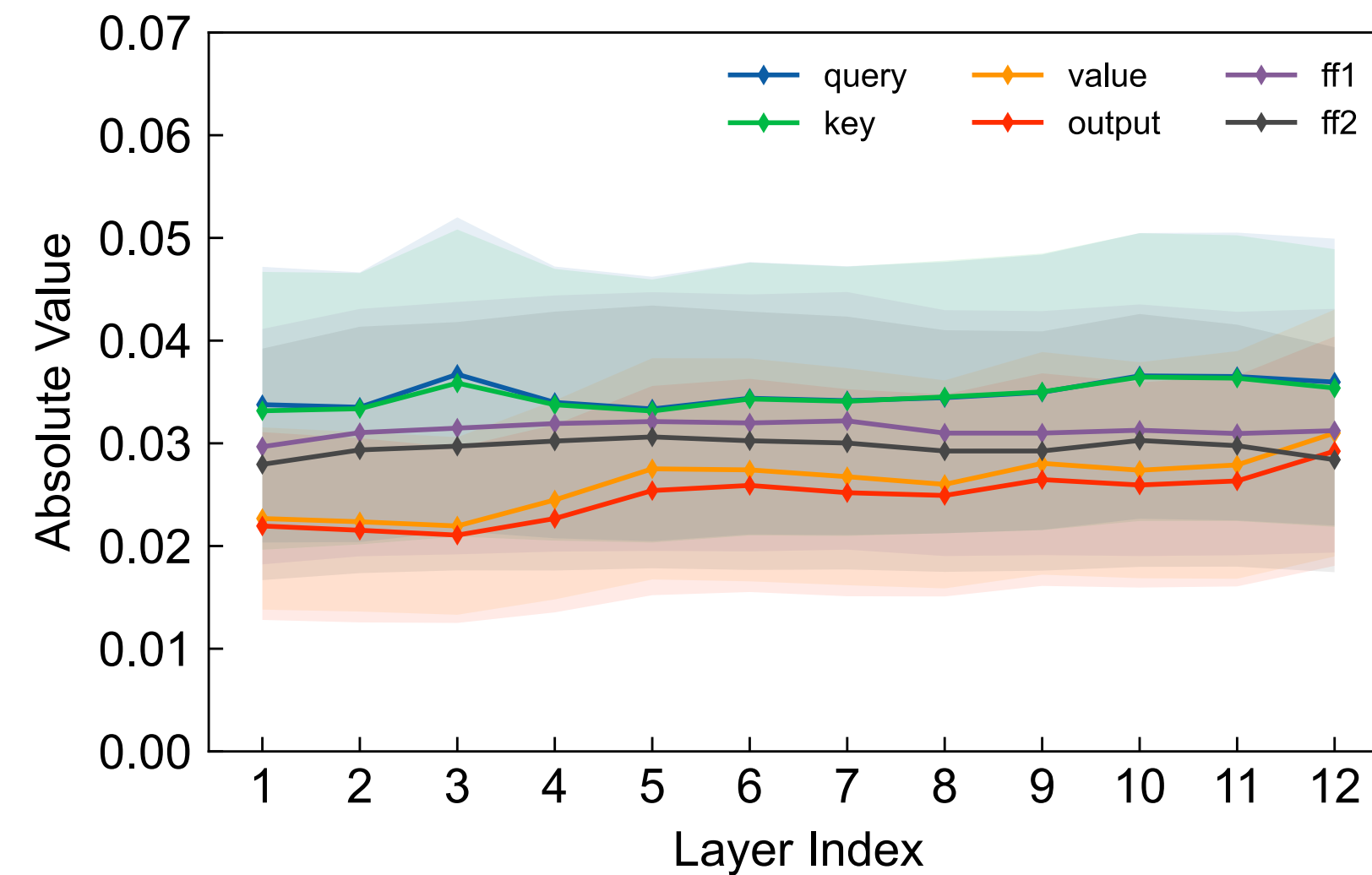
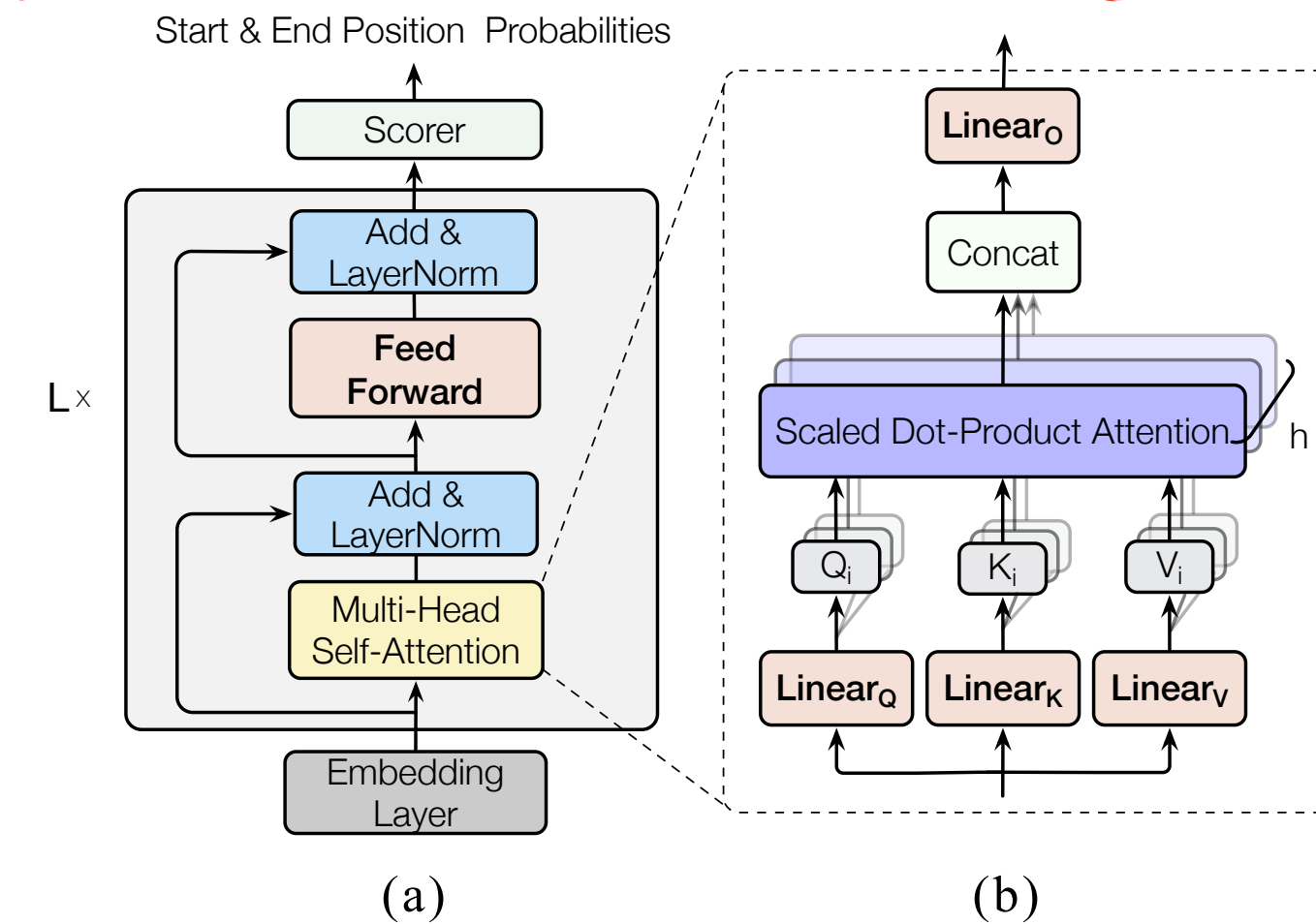
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$$s_n \leftarrow s - s(1 - \frac{n}{N})^2$$



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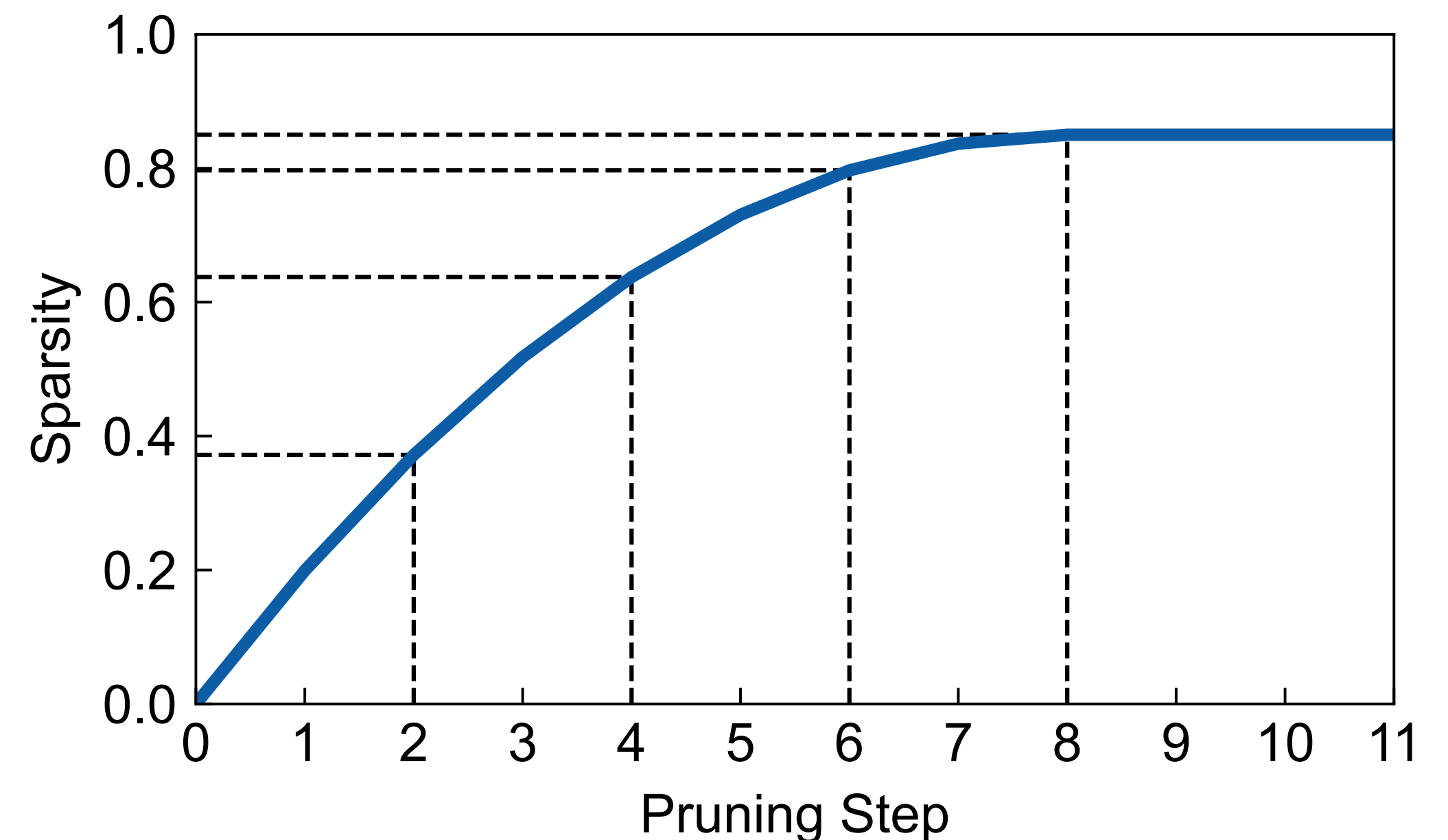
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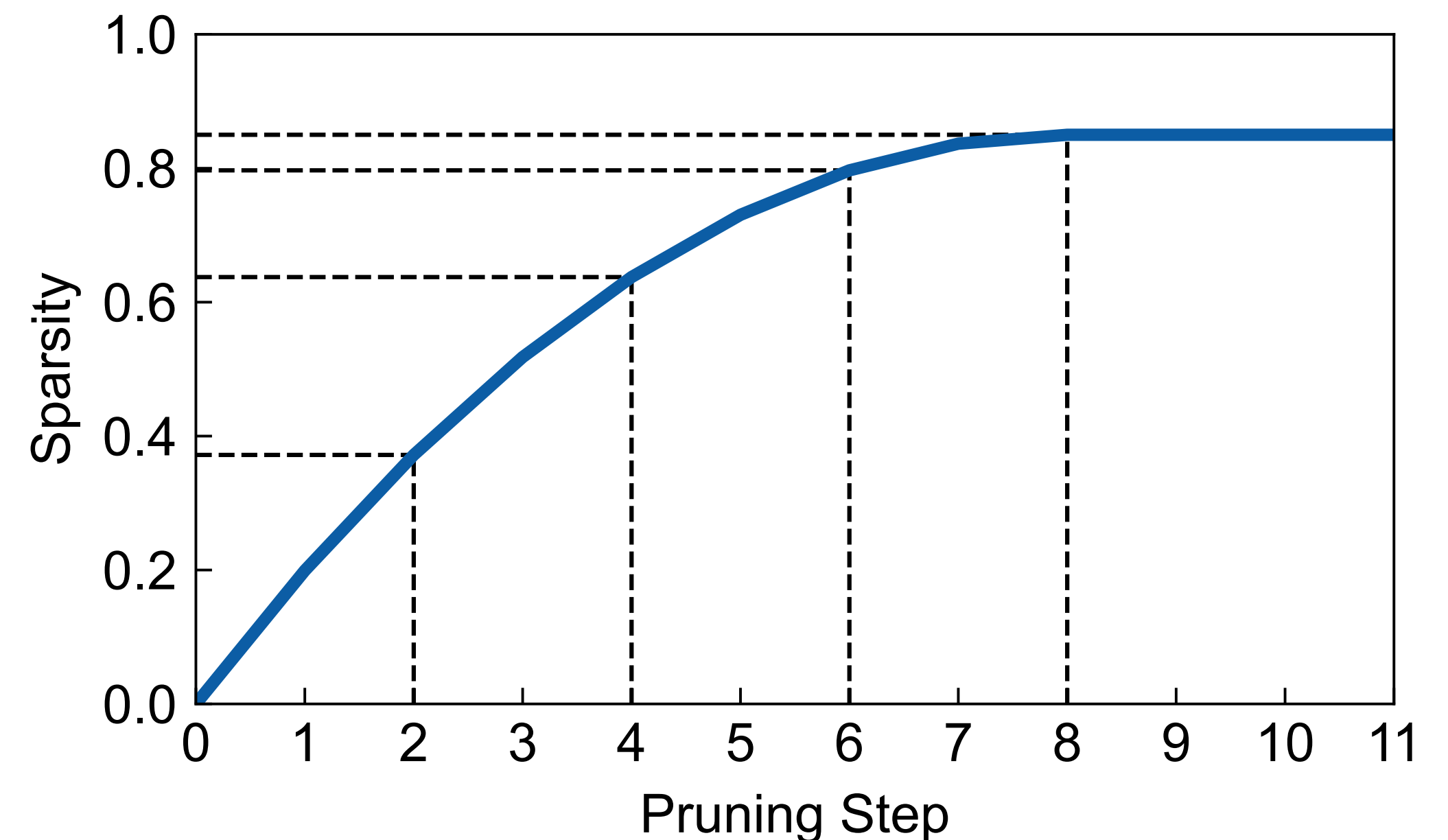
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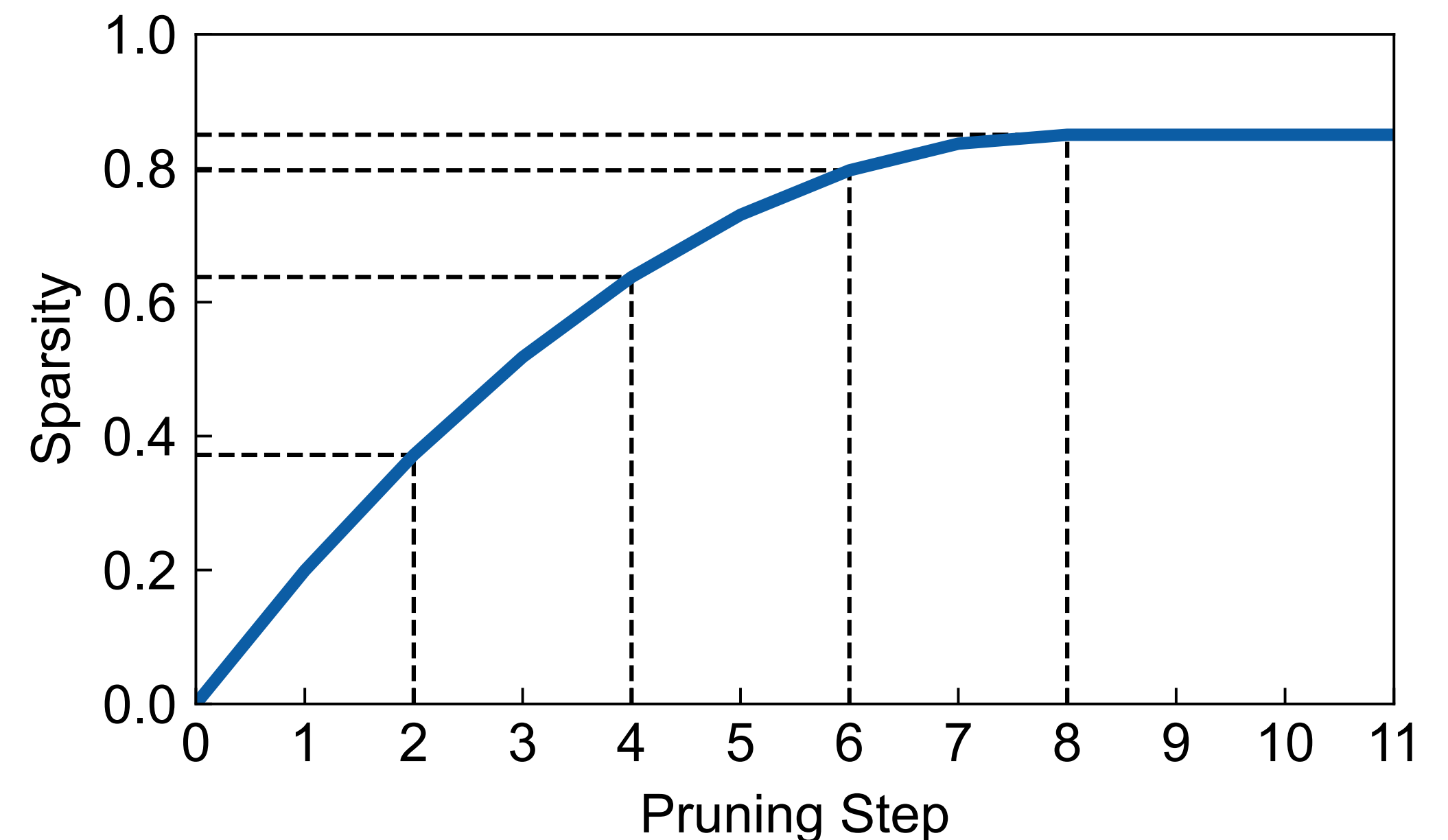
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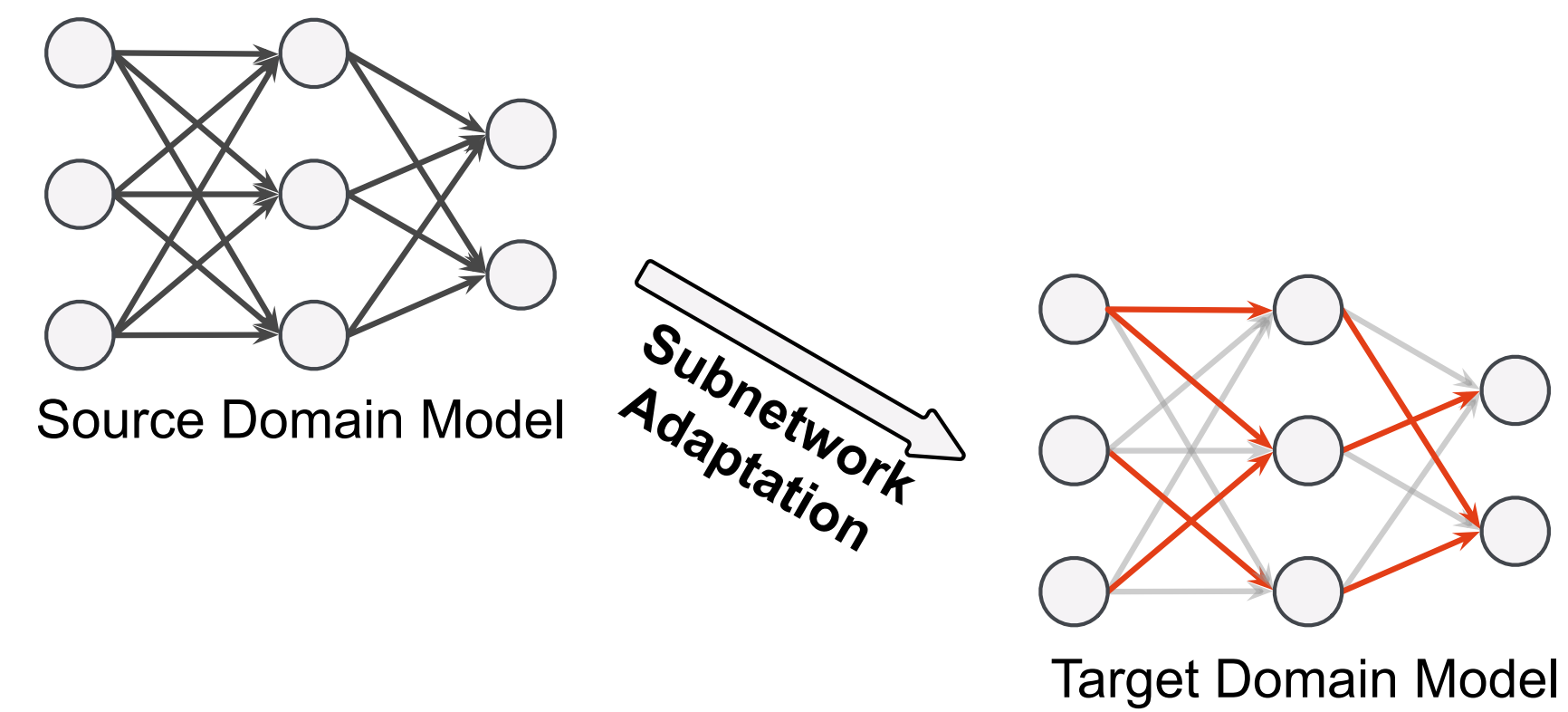
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4. Train the model for ∇t steps
5. Repeat step 1 - 4 N times

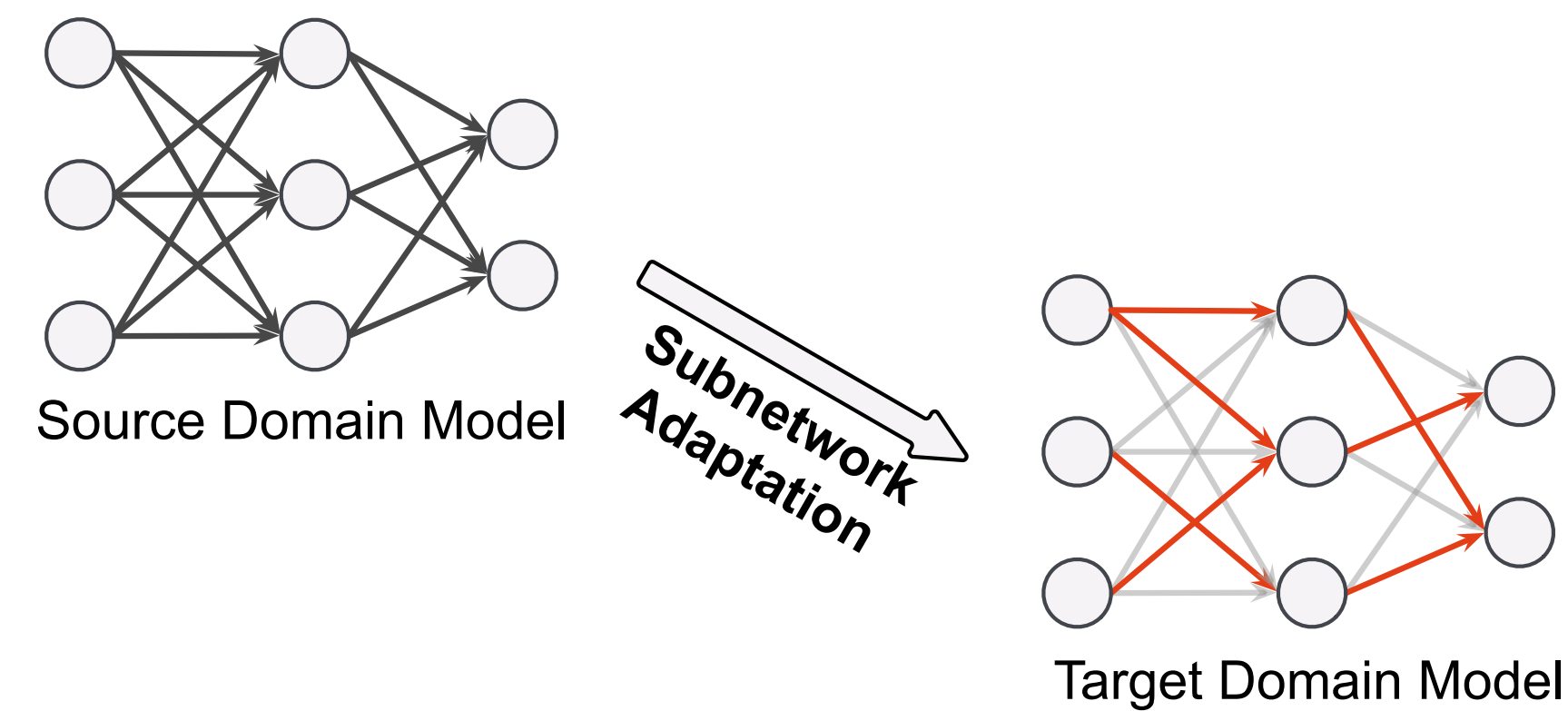


Adapting the Lottery Subnetwork



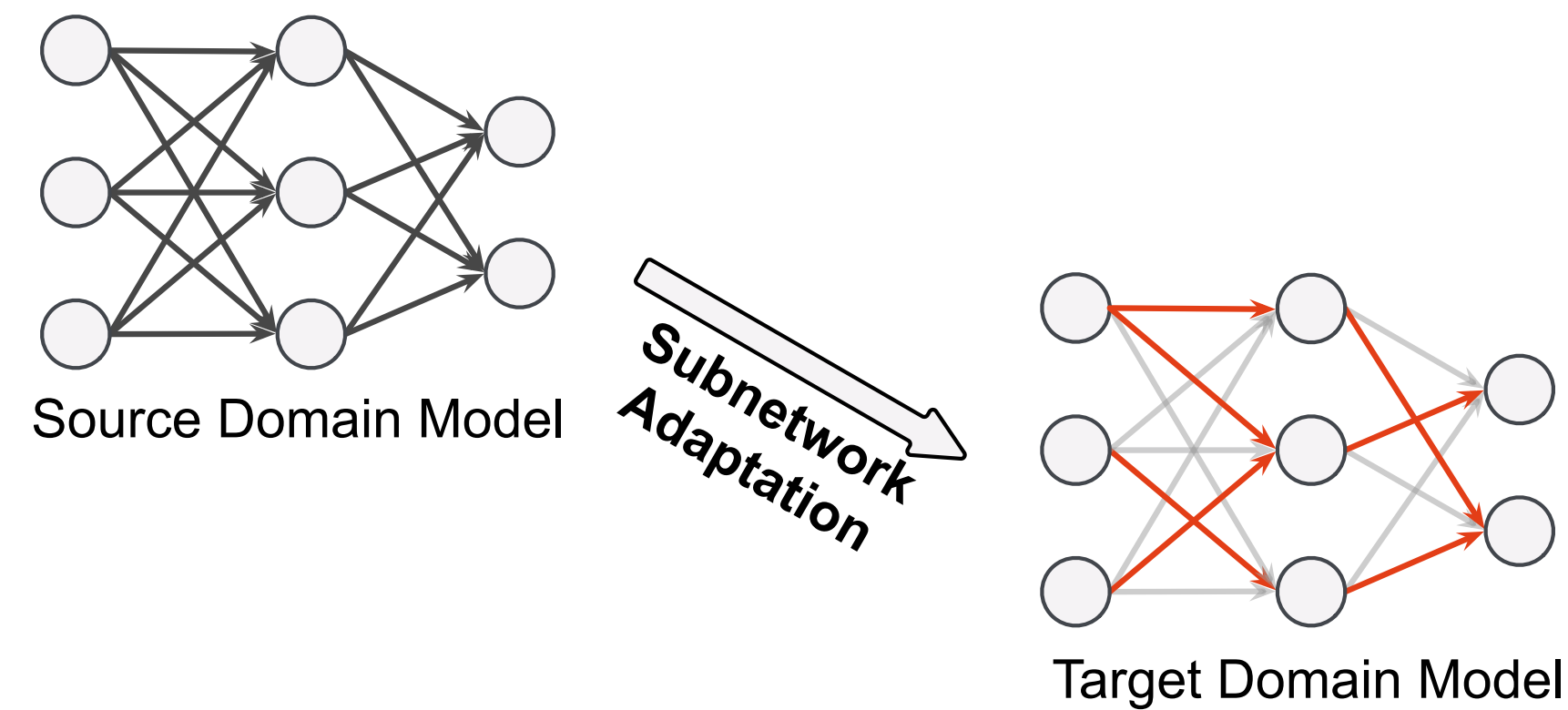
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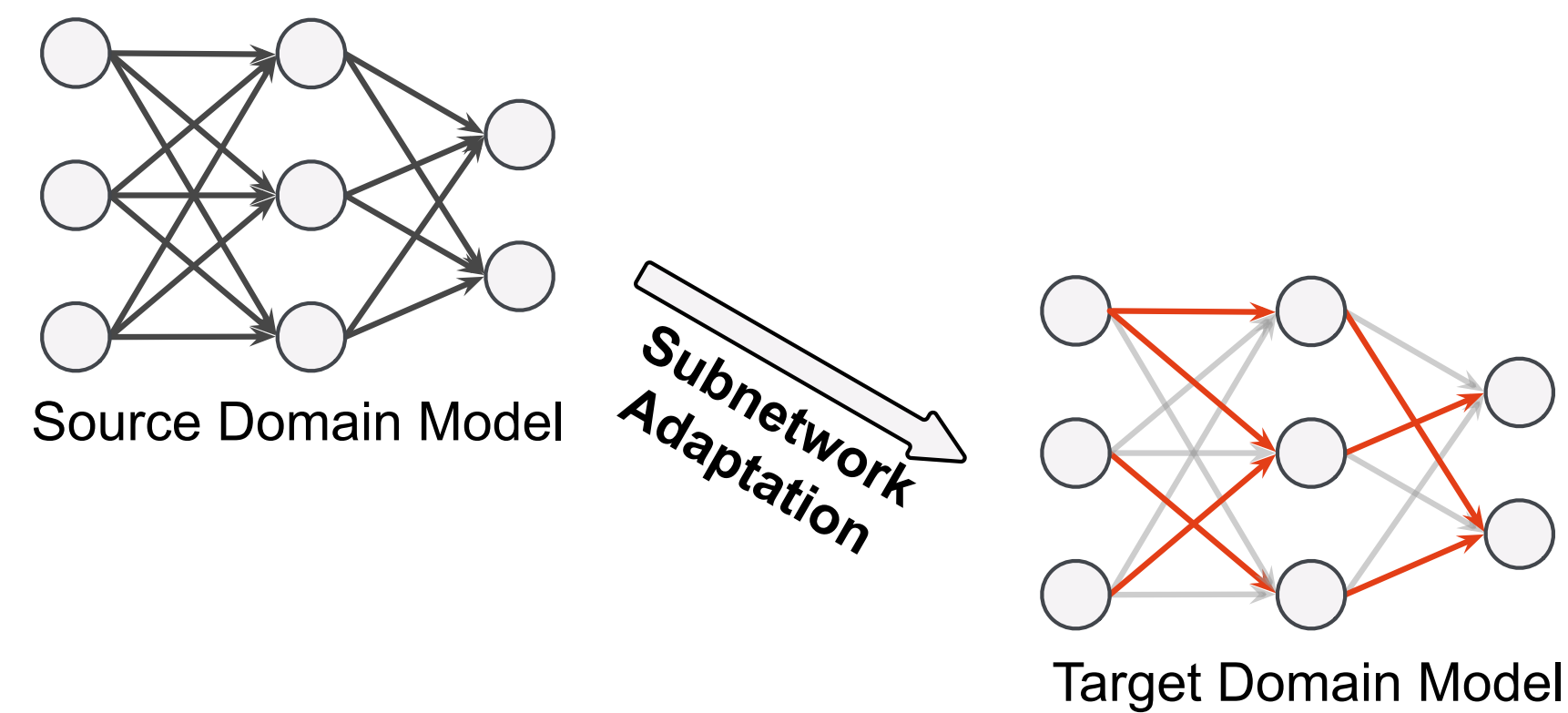
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Adapting the Lottery Subnetwork

- Rewind to the original source domain model parameters
- Only update the parameters of the subnetworks
 - Pruned parameters are frozen, but participate in the forward computation



Experimental Settings

- Source domain: SQuAD
- Five target domains

Dataset	Context	Question	Q C	Train	Dev
SQuAD	Wikipedia	Crowd	✗	87,599	10,507
NewsQA	News Articles	Crowd	✓	74,160	4,212
TriviaQA	Web Snippets	Trivia	✓	61,688	7,785
TweetQA	Tweets	Crowd	✗	7,108	883
NQ	Wikipedia	Queries	✓	104,071	12,836
QuAC	Wikipedia	Crowd	✓	51,695	4,368



Experimental Results

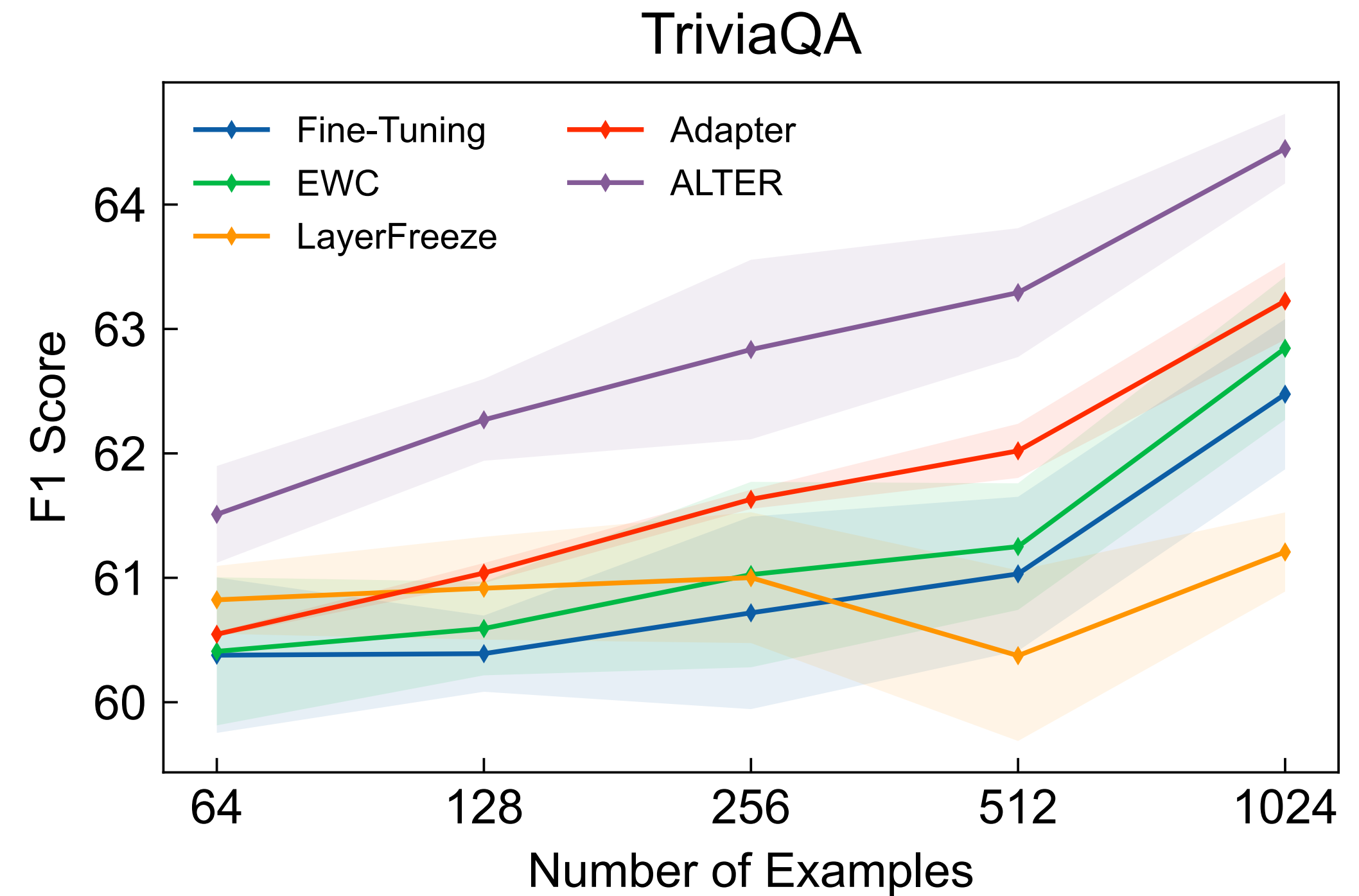
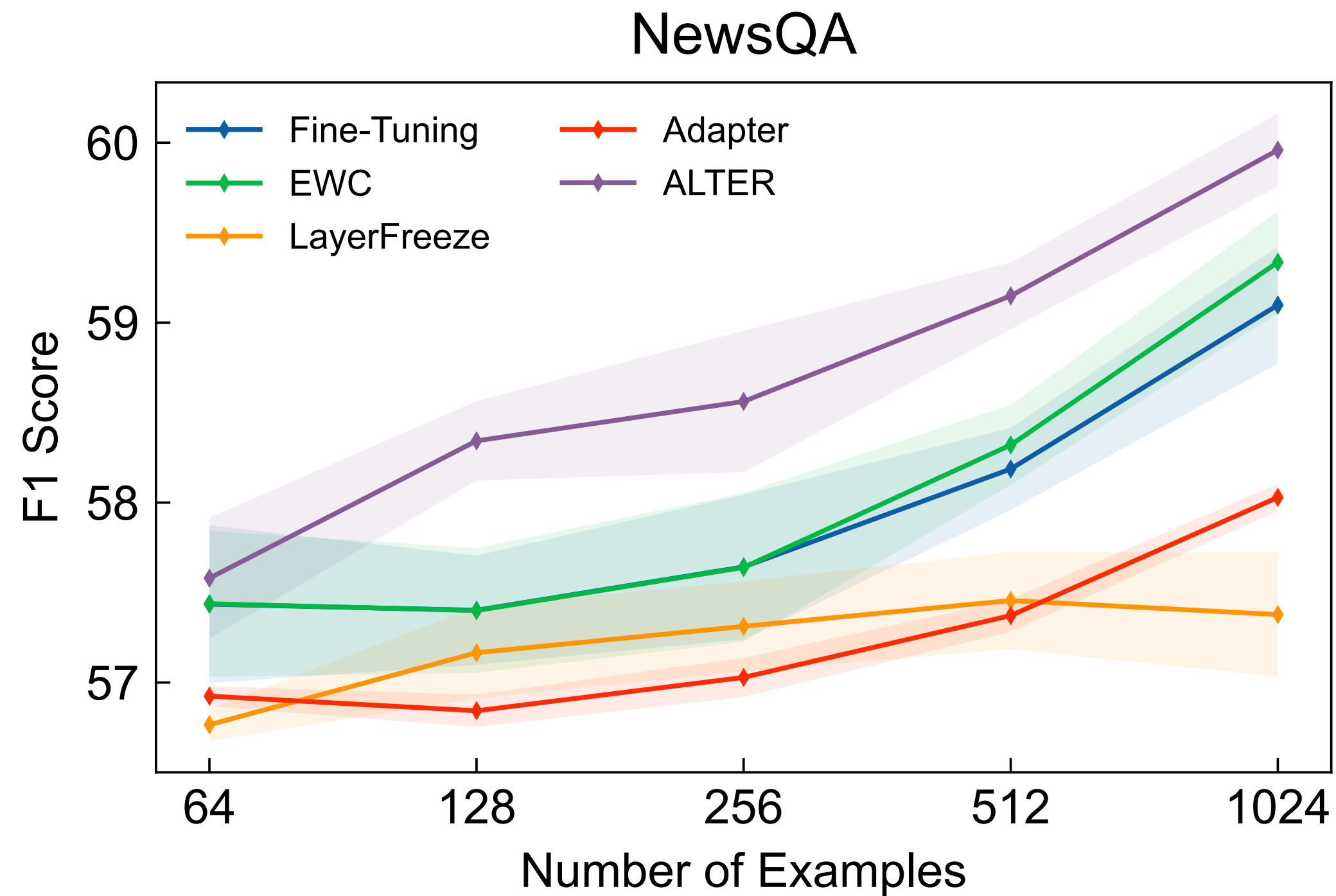
- 1024 target domain examples
- ~20M parameters for adaptation
- Our method **Alter** (**A**daptable **L**ottery)

Model	Training Parameters	NewsQA	TriviaQA	TweetQA	NQ	QuAC
Zero-Shot	None	40.05/56.76	50.52/60.11	67.46/79.48	43,49/57.45	15.82 /37.31
Fine-Tuning	84M	43.24/59.10	55.60/62.48	70.59/81.81	55.23/68.68	26.73/49.25
EWC	84M	43.44/59.34	55.95/62.85	70.48/81.82	55.09/68.54	26.82/49.37
LayerFreeze	21M	40.68/57.38	53.83/61.21	70.32/81.54	50.41/64.11	25.39/47.56
Adapter	20M	41.14/58.03	55.71/63.22	69.50/80.81	49.45/63.44	24.06/46.22
Alter	21M	43.73/59.78	57.47/64.45	71.18/82.31	54.62/68.17	27.50/49.50
Full Data	84M	52.18/66.95	64.44/70.26	68.59/80.58	67.03/78.89	38.37/60.38



Experimental Results

- Varying the number of target domain examples
- 20% ~ 30% parameters are satisfactory



Analysis



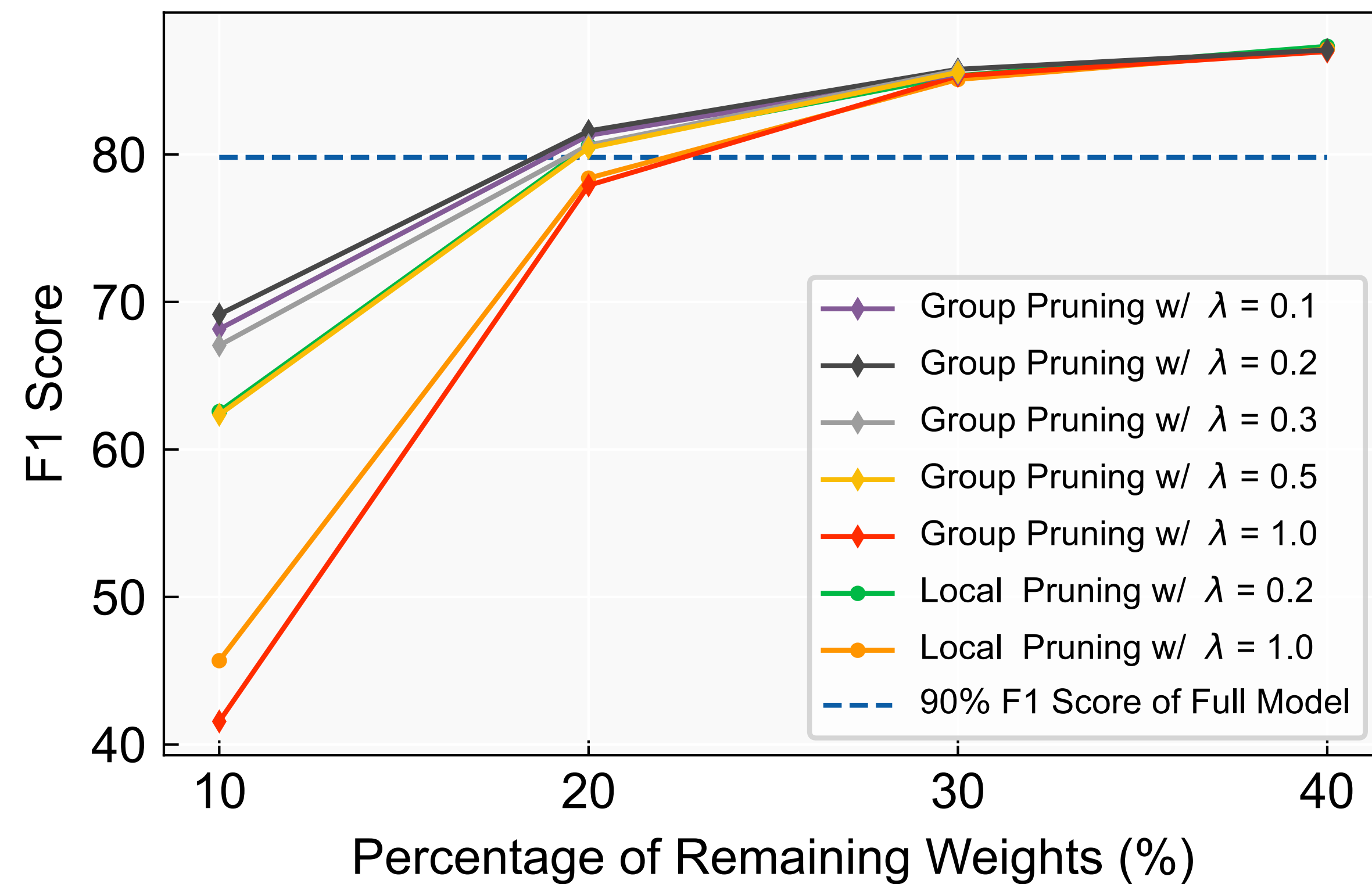
Analysis

- Does structure-aware pruning deliver better lottery subnetworks?



Analysis

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 - Performance on **source domain** by the end of pruning



Analysis



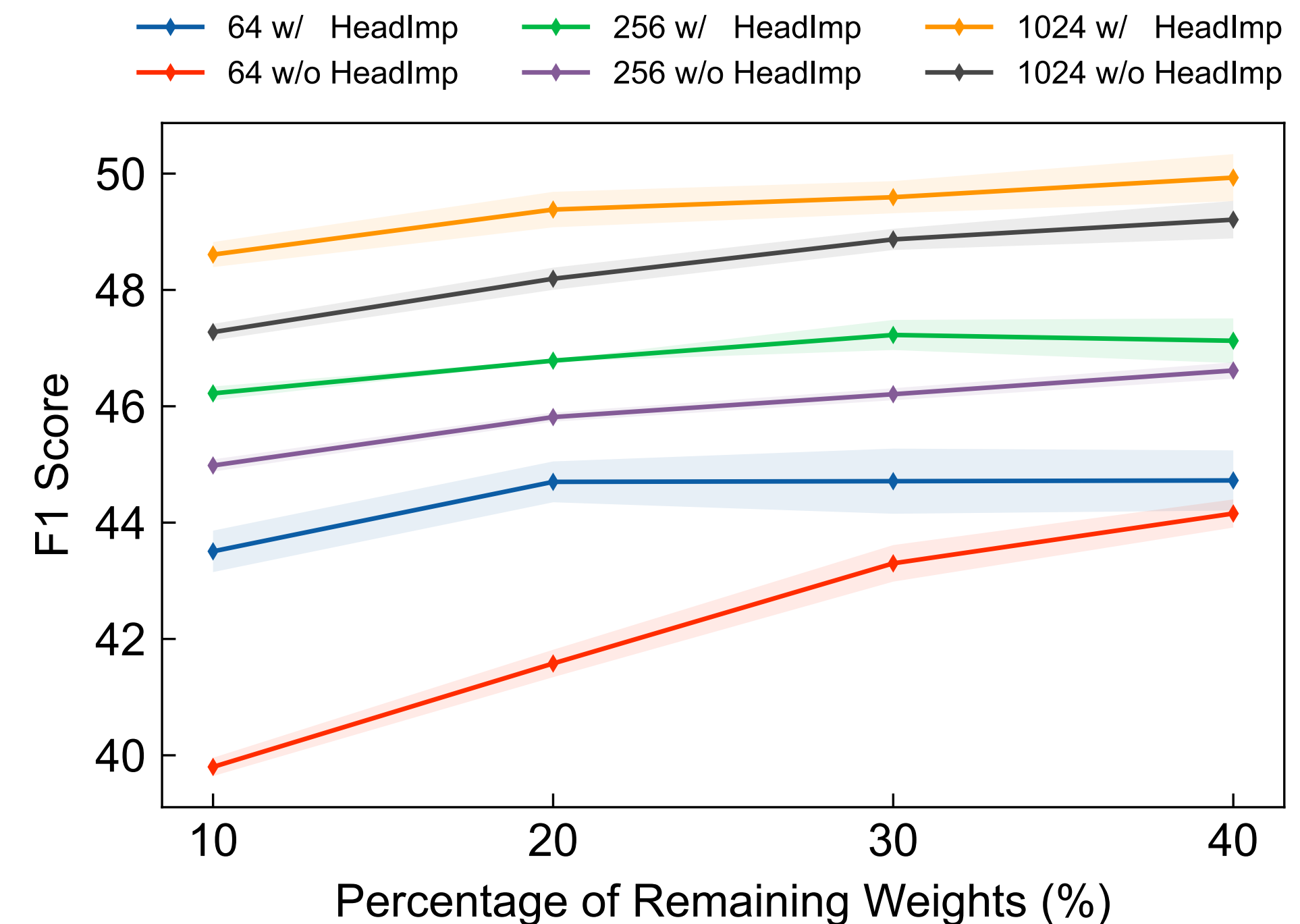
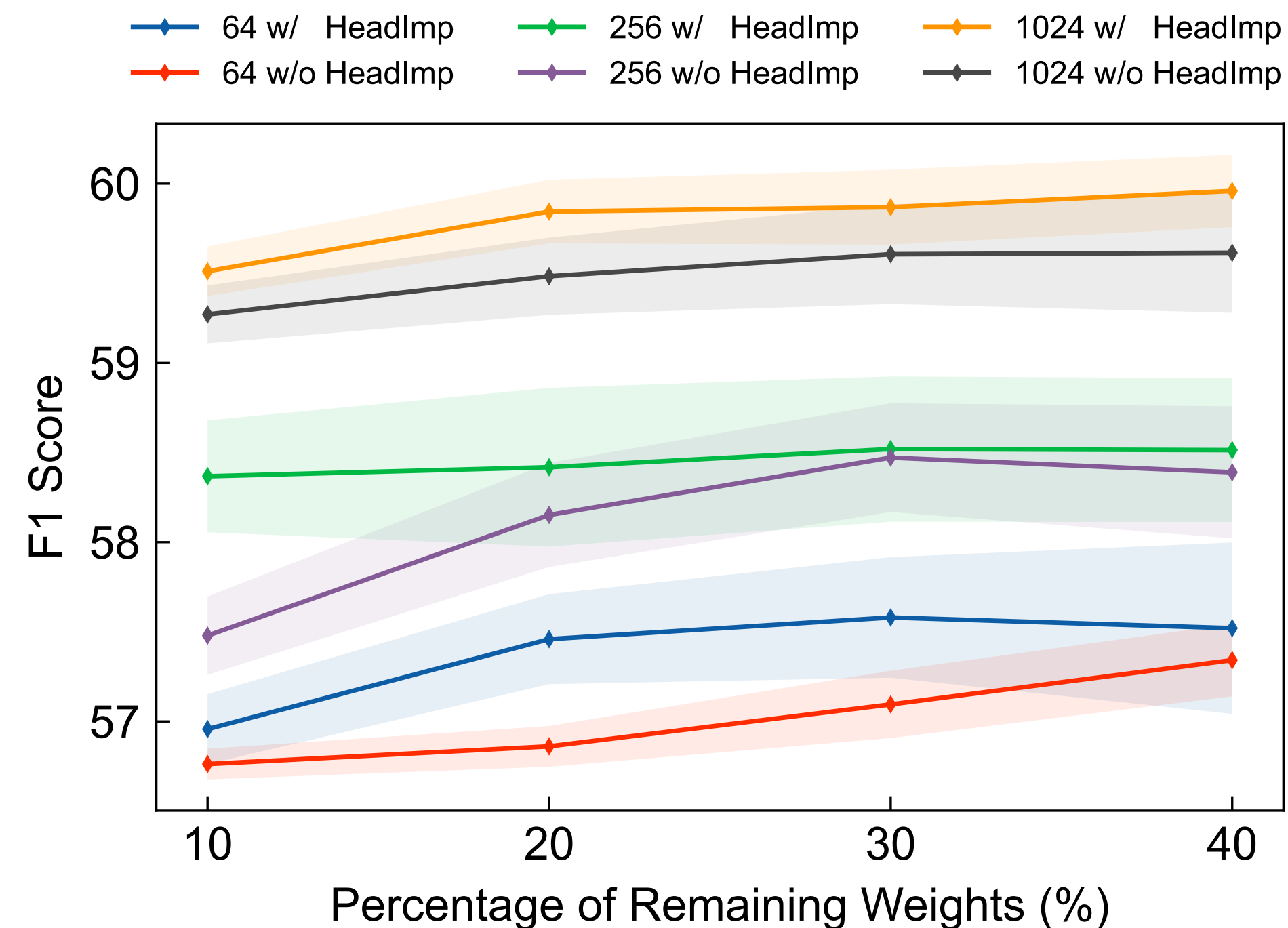
Analysis

- Do better lottery subnetworks improve domain adaptation performance?



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 - Adaptation w/ and w/o **attention head importance** on NewsQA and QuAC



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Method	NewsQA	TriviaQA	TweetQA
Fine-Tuning	40.59/57.40	53.13/60.39	68.23/79.93
Random	40.98/57.72	54.45/61.98	68.57/80.24
Magnitude	40.76/57.56	54.10/62.11	68.76/80.21
Salvage	40.86/57.67	54.39/61.75	68.82/80.24
AttrHead	41.31/58.08	54.35/61.80	68.88/ 80.39
Alter	41.38/58.11	54.60/62.21	68.89/80.35



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- Exploit **a small fraction of parameters** of the over-parameterized source domain model to adapt to the target domain
- Introduce **self-attention attribution** to identify better subnetworks and improve the target domain performance
- Find **subnetwork structures** are critical to the effectiveness besides using fewer parameters



Thanks!

Q&A

The code is publicly available at <https://github.com/haichao592/ALTER>.

